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Chapter 1

Setup, Haar Measure, and Hilbert-Space Framework

1.1 Standing hypotheses

Definition 1 (Standing hypotheses for compact Peter–Weyl). Throughout the blueprint, G denotes a Hausdorff compact topological group, written multiplicatively, with identity 1_G . In Lean we use

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[Group G] [TopologicalSpace G] [IsTopologicalGroup G] [CompactSpace G] [T2Space G] [MeasurableSpace G] [BorelSpace G].
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The scalar field is $k = \mathbb{C}$ (more precisely `[RCLike k]` so the same blueprint covers the real case after the obvious complexification step). The Haar measure is denoted μ and is normalised to be a probability measure ($\mu(G) = 1$).

1.2 Haar measure on a compact group

Theorem 2 (Existence and uniqueness of Haar measure). *Every locally compact Hausdorff topological group G admits a left-invariant regular Borel measure that is finite on compact sets and positive on non-empty open sets, unique up to a positive scalar. In Lean this is the constant `MeasureTheory.Measure.haarMeasure` (the file `Mathlib.MeasureTheory.Measure.Haar.Basic`); the typeclass `IsHaarMeasure` μ packages all the invariance and regularity conditions; uniqueness is in `Mathlib.MeasureTheory.Measure.Haar.Unique` via `haarScalarFactor` and `isMulLeftInvariant_eq_smul_o`*

Proof. Direct citation of the existence and uniqueness theorems in Mathlib’s Haar files. □

Corollary 3 (Haar measure is finite for compact G). *Since G is compact and μ is finite on compact sets, $\mu(G) < \infty$. We may rescale so that $\mu(G) = 1$ (probability normalisation), and we do so throughout.*

Proof. `IsFiniteMeasureOnCompacts` is part of the `IsHaarMeasure` bundle. Rescale via `MeasureTheory.Measure.haar`. □

Theorem 4 (Right-invariance of Haar measure on compact G). *On a compact (Hausdorff) topological group, every left Haar measure is also right-invariant: $\mu(Sg) = \mu(S)$ for every measurable S and $g \in G$. (Equivalently, compact groups are unimodular.)*

Proof. Mathlib's `IsHaarMeasure.IsMulRightInvariant` (or the `IsHaarMeasure.unimodular`-style lemma) handles this. The classical proof: the modular function $\Delta : G \rightarrow (0, \infty)$ is a continuous group homomorphism; on a compact group its image is a compact subgroup of $(0, \infty)$ which is just $\{1\}$, so $\Delta \equiv 1$. $\square$

Theorem 5 (Inversion-invariance of Haar measure). *Pushforward of μ under $g \mapsto g^{-1}$ equals μ .*

Proof. `Measure.inv_eq_self_of_isHaarMeasure` (or the corresponding Mathlib name for unimodular groups). $\square$

1.3 The Hilbert space $L^2(G)$

Definition 6 ($L^2(G)$). $L^2(G) := L^p(\mathbb{C}; \mu)$, the Lebesgue L^2 space of $\mathbb{C}$ -valued square-integrable functions on G modulo a.e. equality, equipped with the inner product $\langle f, h \rangle := \int_G f(x) \overline{h(x)} d\mu(x)$. In Lean this is from `Mathlib.MeasureTheory.Function.L2Space`; the typeclass instance `L2.innerProductSpace` provides the inner product, and `MeasureTheory.Lp.completeSpace` provides completeness.

Theorem 7 ($C(G)$ is dense in $L^2(G)$). *Continuous functions $C(G) \hookrightarrow L^2(G)$ form a dense subspace.*

Proof. Mathlib's general `MeasureTheory.Lp.continuousMap_dense` (via mollification / outer regularity); valid for any finite regular Borel measure on a Polish space, hence in particular for compact Hausdorff G . $\square$

Theorem 8 ($C(G) \subseteq L^\infty(G) \subseteq L^2(G) \subseteq L^1(G)$). *Since μ is finite, L^p inclusions go the natural way: $\|f\|_{L^p} \leq \mu(G)^{1/p-1/q} \|f\|_{L^q}$ for $p \leq q$. Continuous functions on a compact G are bounded so are in L^∞ .*

Proof. Mathlib's `MeasureTheory.MemLp.mono_exponent_of_isFiniteMeasure`. $\square$

Chapter 2

Continuous Unitary Representations

Mathlib's `Representation` `k G V` is purely algebraic: there is no continuity assumption and no Hilbert-space structure. We must add a layer of definitions for the Peter–Weyl theorem. All items in this chapter are [NEW] unless flagged otherwise.

Definition 9 (Continuous unitary representation). A *continuous unitary representation* of G on a (complex) Hilbert space H is a group homomorphism $\rho : G \rightarrow U(H)$ to the unitary group of H , that is continuous when $U(H)$ is given the strong operator topology. In Lean we encode this as a structure `UnitaryRep` `ℂ G H` bundling `toMonoidHom : G →* (H →L[ℂ] H)`, an `isUnitary` field, and a `continuous` field with respect to the strong operator topology on $H \rightarrow L[\mathbb{C}] H$.

Remark 1. *Equivalently, by a standard argument, strong operator continuity of ρ is implied by either of: (a) continuity of $g \mapsto \rho(g)v$ for every $v \in H$; (b) continuity of $(g, v) \mapsto \rho(g)v$. We will use (a) as our default working definition; see Lemma 10.*

Lemma 10 (Strong vs weak operator continuity). *For a homomorphism $\rho : G \rightarrow U(H)$, the following are equivalent:*

1. ρ is continuous in the strong operator topology;
2. for every $v \in H$ the orbit map $g \mapsto \rho(g)v$ is continuous;
3. the action map $G \times H \rightarrow H$, $(g, v) \mapsto \rho(g)v$ is jointly continuous.

Proof. Standard. (1) $\Leftrightarrow$ (2) is the definition of the strong operator topology. (2) $\Rightarrow$ (3) uses uniform boundedness of $\|\rho(g)\| = 1$ together with the $\varepsilon/3$ argument. (3) $\Rightarrow$ (2) is obvious. $\square$

Definition 11 (The left regular representation λ). The *left regular representation* $\lambda : G \rightarrow U(L^2(G))$ is defined by $(\lambda(g)f)(x) := f(g^{-1}x)$ on a continuous representative, extended by continuity to all of $L^2(G)$.

Theorem 12 (λ is a continuous unitary representation). $\lambda(g)$ is unitary on $L^2(G)$ for each g (by left-invariance of μ). The map $g \mapsto \lambda(g)$ is strongly continuous (using density of $C(G)$ in $L^2(G)$ together with uniform continuity of any continuous function on a compact group).

Proof. Unitarity: change of variables under left translation preserves μ by Theorem 2. Strong continuity: for $f \in C(G)$, the orbit map $g \mapsto \lambda(g)f$ is continuous in L^2 by uniform continuity of f on the compact group G (Heine–Cantor); for general $f \in L^2$, use density of $C(G)$ from Theorem 7 and the uniform bound $\|\lambda(g)\| = 1$ via an $\varepsilon/3$ -argument. $\square$

Definition 13 (Right regular representation). $(\rho(g)f)(x) := f(xg)$. Same proofs as for λ show ρ is a continuous unitary representation. The two representations *commute*: $\lambda(g)\rho(h) = \rho(h)\lambda(g)$ for all $g, h \in G$.

Definition 14 (Subrepresentation). A *subrepresentation* of a unitary representation (ρ, H) is a closed ρ -invariant subspace $V \leq H$. Subrepresentations are closed under taking orthogonal complements (since unitary operators preserve orthogonality and closed subspaces).

Lemma 15 (Orthogonal complement of a subrepresentation). *If V is a closed ρ -invariant subspace, then $V^\perp$ is also ρ -invariant. Hence $H = V \oplus V^\perp$ as a direct sum of subrepresentations.*

Proof. $\rho(g)$ is unitary, so it preserves the inner product: if $w \in V^\perp$ and $v \in V$, then $\langle \rho(g)w, v \rangle = \langle w, \rho(g^{-1})v \rangle = 0$ since $\rho(g^{-1})v \in V$. Thus $\rho(g)w \in V^\perp$. $\square$

Definition 16 (Irreducible representation). A continuous unitary representation (ρ, H) is *irreducible* if $H \neq 0$ and the only closed ρ -invariant subspaces are $\{0\}$ and H itself.

Definition 17 (Intertwiner). An *intertwiner* between two continuous unitary representations (ρ, H) and (σ, K) is a bounded linear map $T : H \rightarrow K$ with $T\rho(g) = \sigma(g)T$ for all g . We write $\text{Hom}_G(\rho, \sigma)$ for the space of such intertwiners.

Definition 18 (Equivalence of representations). $(\rho, H) \cong (\sigma, K)$ if there exists a unitary intertwiner $H \rightarrow K$.

Chapter 3

Schur's Lemma in the Compact Setting

3.1 Schur's lemma for irreducible unitary representations

Theorem 19 (Schur's lemma, scalar form). *Let (ρ, H) be an irreducible continuous unitary representation of G . Then every bounded G -equivariant operator $T : H \rightarrow H$ is a scalar: $T = \lambda \cdot \text{Id}_H$ for some $\lambda \in \mathbb{C}$.*

Proof. Decompose $T = A + iB$ with $A := (T + T^*)/2$ and $B := (T - T^*)/(2i)$ self-adjoint. Each of A, B commutes with ρ . Apply the spectral theorem to A : every eigenspace $\ker(A - \lambda I)$ is closed and ρ -invariant, so by irreducibility it is $\{0\}$ or H . Hence the spectrum of A is a single point and A is a scalar; similarly for B .

(For the formalisation: this proof needs the bounded-self-adjoint spectral theorem. A cleaner Mathlib-friendly route is to argue via cyclic vectors using only the existence of a non-trivial reducing projection from approximate eigenspaces, but we follow the standard spectral-theorem approach.) $\square$

Corollary 20 (Schur's lemma, distinct irreducibles). *If (ρ, H) and (σ, K) are irreducible and not isomorphic, then $\text{Hom}_G(\rho, \sigma) = 0$.*

Proof. Let $T \in \text{Hom}_G(\rho, \sigma)$. Then $\overline{\ker T}$ is a closed ρ -invariant subspace of H , so it is 0 or H . Similarly $\overline{T(H)}$ is 0 or K . If $T \neq 0$, then $\ker T = 0$ and $\overline{T(H)} = K$. Polar decomposition $T = U|T|$ expresses T as a unitary $U : H \rightarrow K$ times a positive operator $|T| := \sqrt{T^*T}$ on H . By G -equivariance of T , both U and $|T|$ are G -equivariant. The operator $|T| \in \text{Hom}_G(\rho, \rho)$ is then a positive scalar by Theorem 19, and U is a unitary intertwiner $\rho \rightarrow \sigma$, contradicting non-equivalence. $\square$

Chapter 4

Convolution and Compact Self-adjoint Operators

This is the analytic core of Tao’s proof. Mathlib has convolution defined (`Mathlib.Analysis.LConvolution`, additive notation; `Mathlib.Analysis.Convolution`), but most of the specific facts we need below are [NEW] for compact non-abelian groups.

4.1 Convolution on a compact group

Definition 21 (Convolution on a compact group). For $f, g : G \rightarrow \mathbb{C}$ Borel-measurable, the (multiplicative) convolution is

$$(f * g)(x) := \int_G f(y)g(y^{-1}x) d\mu(y).$$

Mathlib has `MeasureTheory.mconvolution` (in `Mathlib.Analysis.Convolution`, set up via a continuous bilinear form L) for general topological groups.

Theorem 22 (Young’s inequality on a compact group). For $\mu(G) = 1$: $\|f * g\|_{L^2} \leq \|f\|_{L^1} \|g\|_{L^2}$ and $\|f * g\|_{L^\infty} \leq \|f\|_{L^2} \|g\|_{L^2}$. Hence $L^2(G) * L^2(G) \subseteq L^\infty(G) \cap L^2(G) \subseteq C(G)$ (the last when $f \in L^2$ and $g \in C(G)$, or via uniform approximation).

Proof. Direct computation; in Mathlib this follows from versions of Young’s inequality in `MeasureTheory.MemLp.convolution` after specialising to compact groups. $\square$

Theorem 23 (Convolution by an L^2 kernel is compact on $L^2(G)$). Let $\phi \in L^2(G)$. Then the operator $T_\phi : L^2(G) \rightarrow L^2(G)$, $f \mapsto f * \phi$, is a Hilbert–Schmidt operator with kernel $K(x, y) = \phi(y^{-1}x) \in L^2(G \times G)$. In particular T_ϕ is a compact operator (in the sense of Mathlib’s `IsCompactOperator`).

Proof. The operator T_ϕ is given by an integral kernel $K(x, y) = \phi(y^{-1}x)$. Compute its Hilbert–Schmidt norm:

$$\int \int |K(x, y)|^2 d\mu d\mu = \int \int |\phi(y^{-1}x)|^2 d\mu(y) d\mu(x) = \mu(G) \int |\phi|^2 = \|\phi\|_{L^2}^2 < \infty,$$

using right-invariance (Theorem 4) and inversion-invariance of μ . Hilbert–Schmidt operators are compact: this requires the (currently unformalised in Mathlib for the general Hilbert-Schmidt

case) fact that integral operators with L^2 kernels are compact. Equivalently: the Hilbert-Schmidt norm bounds the operator norm by the trace-class lemma. $\square$

Theorem 24 (Self-adjointness of convolution). *Let $\phi \in L^2(G)$ satisfy $\phi(g) = \overline{\phi(g^{-1})}$ for μ -a.e. g . Then the operator $T_\phi : f \mapsto f * \phi$ is self-adjoint on $L^2(G)$.*

Proof. Compute $\langle T_\phi f, h \rangle$ and $\langle f, T_\phi h \rangle$ via Fubini. The condition $\phi(g) = \overline{\phi(g^{-1})}$ exactly switches the two integrands after a substitution $g \mapsto g^{-1}$ (using inversion invariance, Theorem 5). $\square$

Theorem 25 (Convolution commutes with right translation). *For every $\phi \in L^2(G)$ and $g \in G$, $T_\phi \circ \rho(g) = \rho(g) \circ T_\phi$.*

Proof. Direct change of variables: for $f \in L^2(G)$, $T_\phi(\rho(g)f)(x) = \int f(yg)\phi(y^{-1}x)d\mu(y) = \int f(z)\phi(z^{-1}gx)d\mu(z)$ (substituting $z = yg$, using Theorem 4), which is $(T_\phi f)(gx) = \rho(g)(T_\phi f)(x)$. $\square$

4.2 Spectral theorem for compact self-adjoint operators

Theorem 26 (Spectral theorem for compact self-adjoint operators). *Let T be a non-zero compact self-adjoint operator on a complex Hilbert space H . Then:*

1. *the spectrum $\sigma(T) \setminus \{0\}$ consists of countably many real eigenvalues $\{\lambda_n\}$ with $\lambda_n \rightarrow 0$;*
2. *each non-zero eigenvalue has finite multiplicity, i.e. the eigenspace $H_{\lambda_n} := \ker(T - \lambda_n I)$ is finite-dimensional;*
3. *we have an orthogonal Hilbert-sum decomposition $H = \ker T \oplus \widehat{\bigoplus_n H_{\lambda_n}}$.*

Equivalently: there exists a (countable) orthonormal basis of H consisting of eigenvectors of T .

Proof. Standard. Sketch: the operator norm $\|T\|$ is attained as $\sup\{|\langle Tx, x \rangle|\} = \pm\lambda_1$, an eigenvalue, with finite-dimensional eigenspace H_{λ_1} ; pass to $H_{\lambda_1}^\perp$ which is T -invariant, and iterate. The iterating eigenvalues form a sequence going to zero by compactness (any bounded set of eigenvectors with eigenvalues bounded below has a convergent subsequence, but eigenvectors of distinct eigenvalues are orthogonal, so a uniformly lower-bounded sequence cannot have a convergent subsequence). The closure of the span of all eigenvectors with non-zero eigenvalue equals $(\ker T)^\perp$.

This is a non-trivial result of functional analysis that, to the author's knowledge, is not currently in Mathlib (as of knowledge cutoff Jan 2026; please verify against current Mathlib state). The finite-dimensional version `LinearMap.IsSymmetric.diagonalization` is in `Mathlib.Analysis.InnerProductSpace` extending to the compact-operator case is itself a substantial formalisation project. $\square$

Chapter 5

Finite-Dimensionality of Eigenspaces and Existence of Irreducibles

5.1 Eigenspaces of G -equivariant compact operators

Theorem 27 (Eigenspaces of equivariant compact operators are G -invariant). *Let $\phi \in L^2(G)$ and T_ϕ the convolution operator on $L^2(G)$. For each $\lambda \in \mathbb{C}$, the eigenspace $H_\lambda := \ker(T_\phi - \lambda I)$ is closed and right-translation-invariant (i.e. a closed subrepresentation of the right regular representation).*

Proof. T_ϕ commutes with $\rho(g)$ (Theorem 25), so if $T_\phi v = \lambda v$ then $T_\phi(\rho(g)v) = \rho(g)T_\phi v = \lambda \rho(g)v$, i.e. $\rho(g)v \in H_\lambda$. $\square$

Corollary 28 (Every irreducible subrepresentation of $L^2(G)$ is finite-dimensional). *Let $V \leq L^2(G)$ be a non-zero irreducible right-translation-invariant closed subspace. Then $\dim V < \infty$.*

Proof. Let $0 \neq v \in V$. Pick $\phi \in C(G)$ self-adjoint ($\phi(g) = \overline{\phi(g^{-1})}$) such that $T_\phi v \neq 0$ (such ϕ exists: e.g. a small bump function around 1_G approximates the identity, see Lemma 29). By Theorem 23 and Theorem 24, T_ϕ is compact and self-adjoint, and by the spectral theorem (Theorem 26) it has at least one non-zero eigenvalue λ , with H_λ finite-dimensional. By Theorem 27 the eigenspace H_λ is right-translation-invariant, so $V \cap H_\lambda$ is a non-zero (since $T_\phi v \neq 0$ for some eigenfunction's spectral component lies in V) right-invariant subspace of V . By irreducibility of V , $V \cap H_\lambda = V$, so $V \subseteq H_\lambda$, which is finite-dimensional. $\square$

Lemma 29 (Approximate identity from continuous bumps). *There exists a net $\{\phi_U\}_{U \in \mathcal{N}(1_G)}$ of continuous non-negative functions $\phi_U : G \rightarrow \mathbb{R}_{\geq 0}$, indexed by open neighbourhoods U of 1_G , satisfying: $\text{supp}(\phi_U) \subseteq U$, $\int \phi_U d\mu = 1$, $\phi_U(g) = \phi_U(g^{-1})$, and $f * \phi_U \rightarrow f$ in L^2 as $U \rightarrow \{1_G\}$ for every $f \in L^2(G)$.*

Proof. Use Urysohn's lemma to obtain continuous non-negative bump functions with support in U , normalise the integral, symmetrise via $\phi_U(g) := \frac{1}{2}(\psi_U(g) + \psi_U(g^{-1}))$. Convergence $f * \phi_U \rightarrow f$ uses uniform continuity of $f \in C(G)$ on the compact group G followed by density of $C(G)$ in L^2 . $\square$

Chapter 6

The Peter–Weyl Decomposition of $L^2(G)$

6.1 Matrix coefficients

Definition 30 (Matrix coefficient). Given a finite-dimensional continuous unitary representation (ρ, V) of G and vectors $u, v \in V$, the function $\pi_{u,v}(g) := \langle \rho(g)u, v \rangle_V$ is the *matrix coefficient* associated to (ρ, u, v) . Matrix coefficients are continuous functions on G and bounded (by $\|u\| \|v\|$). Hence $\pi_{u,v} \in C(G) \cap L^2(G)$.

Lemma 31 (Algebra structure on matrix coefficients). *The set of all matrix coefficients (over all finite-dimensional continuous unitary representations of G) is closed under: pointwise addition (taking direct sums); pointwise multiplication (taking tensor products); complex conjugation (taking the contragredient $\bar{\rho}$); and contains the constants (the trivial representation).*

Proof. Standard verifications: e.g. if $f_1 = \pi_{u_1, v_1}^{(1)}$ for ρ_1 on V_1 and $f_2 = \pi_{u_2, v_2}^{(2)}$ for ρ_2 on V_2 , then $f_1 \cdot f_2 = \pi_{u_1 \otimes u_2, v_1 \otimes v_2}^{(\rho_1 \otimes \rho_2)}$. Conjugation $\overline{f_1}$ is a matrix coefficient of the contragredient representation on V_1^* . $\square$

Definition 32 (The matrix-coefficient subalgebra $\mathcal{A}$). Let $\mathcal{A} := \text{span}_{\mathbb{C}} \{\pi_{u,v} : (\rho, V) \text{ fin-dim cts unitary rep, } u, v \in V\} \subseteq C(G)$. By Lemma 31, $\mathcal{A}$ is a unital $*$ -subalgebra of the (commutative under pointwise multiplication) $*$ -algebra $C(G)$. Equivalently: a star-subalgebra of the Mathlib bundled type `StarSubalgebra  $\mathbb{C} C(G, \mathbb{C})$` .

6.2 $\mathcal{A}$ separates points: existence of irreducibles

This is the key step in proving Peter–Weyl. Tao calls the underlying proposition “every non-trivial element of G acts non-trivially on some finite-dimensional irreducible”.

Theorem 33 (Existence of separating finite-dimensional irreducibles). *For every $g \in G$ with $g \neq 1_G$, there exists a finite-dimensional continuous unitary irreducible representation (ρ, V) of G with $\rho(g) \neq \text{Id}_V$.*

Proof. Following Tao. Pick $g \neq 1_G$. By compactness and Hausdorffness, pick disjoint open neighbourhoods $U \ni 1_G$ and $V \ni g$, and a continuous symmetric non-negative bump ϕ supported in U with $\int \phi = 1$.

Then $T_\phi(\mathbf{1}_V) = \mathbf{1}_V * \phi$ is supported in $V \cdot U$ (which is disjoint from a smaller neighbourhood of $\mathbf{1}_G$), so the function $T_\phi(\mathbf{1}_V) - \mathbf{1}_V$ is non-zero. Pick a non-zero eigenvalue λ of T_ϕ such that H_λ is not contained in the g -fixed subspace of $L^2(G)$ (this is possible since the spans of all H_λ are dense in $(\ker T_\phi)^\perp$ by Theorem 26, and g acts non-trivially on $L^2(G)$ via the right regular representation $\rho(g)$, since $\rho(g)\mathbf{1}_V \neq \mathbf{1}_V$ when V is small).

By Theorem 27 the eigenspace H_λ is finite-dimensional and right- G -invariant. Decompose H_λ into irreducible subrepresentations (possible since H_λ is a finite-dimensional unitary representation, which is semisimple by Lemma 15). At least one irreducible component $V \subseteq H_\lambda$ has $\rho(g) \neq \text{Id}_V$ (else g acts trivially on H_λ , contradicting the choice). $\square$

Corollary 34 ($\mathcal{A}$ separates points). *The matrix-coefficient $*$ -subalgebra $\mathcal{A} \subseteq C(G, \mathbb{C})$ separates points of G .*

Proof. Let $g \neq h$ in G . Apply Theorem 33 to gh^{-1} to obtain a finite-dimensional irreducible (ρ, V) with $\rho(gh^{-1}) \neq \text{Id}_V$. Pick $u, v \in V$ with $\langle \rho(gh^{-1})u, v \rangle \neq \langle u, v \rangle$, i.e. $\pi_{u,v}(gh^{-1}) \neq \pi_{u,v}(1_G)$. Right-translating by h gives $\pi_{u,v}(g) \neq \pi_{\rho(h)u,v}(h) = \pi_{u,v}(h)$ via the matrix-coefficient identity $\pi_{u,v}(gh) = \langle \rho(g)\rho(h)u, v \rangle = \pi_{\rho(h)u,v}(g)$. $\square$

6.3 Density of $\mathcal{A}$ via Stone–Weierstrass

Theorem 35 ($\mathcal{A}$ is dense in $C(G)$). *The closure of $\mathcal{A}$ in the uniform topology of $C(G, \mathbb{C})$ is all of $C(G, \mathbb{C})$.*

Proof. Apply the complex Stone–Weierstrass theorem `ContinuousMap.starSubalgebra_topologicalClosure_eq_top` (`Mathlib.Topology.ContinuousMap.StoneWeierstrass`) to $\mathcal{A}$. Hypotheses: G compact (Definition 1); $\mathcal{A}$ a star-subalgebra of $C(G, \mathbb{C})$ (Definition 32); $\mathcal{A}$ separates points (Corollary 34). Conclusion: $\mathcal{A}$ is dense in $C(G)$ in the uniform topology. $\square$

Corollary 36 ($\mathcal{A}$ is dense in $L^2(G)$). *The closure of $\mathcal{A}$ in $L^2(G)$ (under the L^2 topology) is all of $L^2(G)$.*

Proof. $\mathcal{A}$ is uniformly dense in $C(G)$ by Theorem 35, hence L^2 -dense in $C(G)$ since $\|\cdot\|_{L^2} \leq \|\cdot\|_{L^\infty}$ on a probability space (Theorem 8). And $C(G)$ is L^2 -dense in $L^2(G)$ (Theorem 7). $\square$

6.4 The isotypic decomposition of $L^2(G)$

Definition 37 (Isotypic component). Given an irreducible class $[\xi]$ (an isomorphism class of finite-dimensional continuous irreducible unitary representations of G), the ξ -*isotypic component* of $L^2(G)$ (with respect to the right regular representation ρ) is $L^2(G)_\xi := \overline{\sum_{V \cong \xi} V}$, the closure of the sum of all closed right- G -invariant subspaces of $L^2(G)$ that are isomorphic to ξ .

Lemma 38 (Isotypic components are pairwise orthogonal). *$L^2(G)_\xi \perp L^2(G)_\eta$ whenever $[\xi] \neq [\eta]$.*

Proof. Pick $V \cong \xi$ and $W \cong \eta$ closed irreducible subrepresentations of $L^2(G)$. The orthogonal projection $P_W : L^2(G) \rightarrow W$ is bounded and G -equivariant (Lemma 15); restricting to V gives an intertwiner $V \rightarrow W$, which is zero by Schur (Corollary 20). Hence $V \perp W$. Since closures preserve orthogonality, the same holds for $L^2(G)_\xi$ and $L^2(G)_\eta$. $\square$

Theorem 39 (Peter–Weyl: L^2 decomposition). *$L^2(G)$ is the orthogonal Hilbert sum of the ξ -isotypic components, as $[\xi]$ ranges over all isomorphism classes of finite-dimensional continuous unitary irreducible representations of G :*

$$L^2(G) = \widehat{\bigoplus_{[\xi]} L^2(G)_\xi}.$$

In particular every $f \in L^2(G)$ has a unique decomposition $f = \sum_{[\xi]} f_\xi$ with $f_\xi \in L^2(G)_\xi$, with $\sum \|f_\xi\|_2^2 = \|f\|_2^2$ (Parseval–Plancherel).

Proof. By Lemma 38 the family $\{L^2(G)_\xi\}_{[\xi]}$ is an orthogonal family in $L^2(G)$. The closed sum $\overline{\sum_{[\xi]} L^2(G)_\xi}$ is a closed G -invariant subspace of $L^2(G)$. Each matrix coefficient $\pi_{u,v}$ of an irreducible ξ lies in $L^2(G)_\xi$ (by the matrix-coefficient construction), so $\mathcal{A}$ is contained in the closed sum. By Corollary 36, $\mathcal{A}$ is dense in $L^2(G)$, so the closed sum equals $L^2(G)$.

In Lean: this is exactly the predicate `IsHilbertSum` from `Mathlib.Analysis.InnerProductSpace.l2Space`, applied to the orthogonal family of inclusion isometries $L^2(G)_\xi \hookrightarrow L^2(G)$. $\square$

Theorem 40 (Peter–Weyl: explicit form on each isotypic component). *Each isotypic component $L^2(G)_\xi$ is itself an orthogonal Hilbert sum of $\dim V_\xi$ -many copies of V_ξ (so $L^2(G)_\xi$ has $\mathbb{C}$ -dimension $(\dim V_\xi)^2$ when $\dim V_\xi < \infty$).*

Proof. Standard. Use the (left, right) bi-equivariance of $L^2(G)$: $L^2(G)_\xi$ is also a closed left- G -invariant subspace, and as a representation under the joint $G \times G$ -action it is isomorphic to $V_\xi \otimes V_\xi^*$. The matrix coefficient map $V_\xi \otimes V_\xi^* \rightarrow L^2(G)$, $u \otimes \ell \mapsto (g \mapsto \ell(\rho_\xi(g)^{-1}u))$, is an injective $G \times G$ intertwiner with image $L^2(G)_\xi$; orthogonality $\langle u_1 \otimes \ell_1, u_2 \otimes \ell_2 \rangle = (\dim V_\xi)^{-1} \langle u_1, u_2 \rangle \overline{\langle \ell_1, \ell_2 \rangle}$ follows from Schur’s orthogonality relations. $\square$

Chapter 7

Schur Orthogonality, Plancherel, and Fourier Inversion

7.1 Schur orthogonality of matrix coefficients

Theorem 41 (Schur orthogonality, distinct irreducibles). *Let (ρ, V) , (σ, W) be finite-dimensional continuous unitary irreducibles with $\rho \not\cong \sigma$. Then for every $u_1, u_2 \in V$ and $w_1, w_2 \in W$,*

$$\int_G \langle \rho(g)u_1, u_2 \rangle_V \cdot \overline{\langle \sigma(g)w_1, w_2 \rangle_W} d\mu(g) = 0.$$

Equivalently $\pi_{u_1, u_2}^\rho \perp \pi_{w_1, w_2}^\sigma$ in $L^2(G)$.

Proof. Direct from Lemma 38: matrix coefficients of ρ live in the $[\rho]$ -isotypic component of $L^2(G)$, and those of σ in the $[\sigma]$ -component, which are orthogonal. $\square$

Theorem 42 (Schur orthogonality, same irreducible). *Let (ρ, V) be a finite-dimensional continuous unitary irreducible with $d := \dim V$. Then for $u_1, u_2, v_1, v_2 \in V$,*

$$\int_G \langle \rho(g)u_1, v_1 \rangle \cdot \overline{\langle \rho(g)u_2, v_2 \rangle} d\mu(g) = \frac{1}{d} \langle u_1, u_2 \rangle \cdot \overline{\langle v_1, v_2 \rangle}.$$

Proof. Standard. Define a G -equivariant operator $T : V \rightarrow V$, $Tu := \int_G \rho(g)u \otimes \overline{\rho(g)v_1} \cdot \pi_{u_2, v_2}(g)^* d\mu$ (compactness of G ensures convergence). Use Theorem 19 to identify T with a scalar, and compute the scalar by taking traces. $\square$

Corollary 43 (Orthonormal basis from matrix coefficients). *Let $\widehat{G}$ index a choice of one finite-dimensional unitary irreducible V_ξ from each isomorphism class. For each ξ , fix an orthonormal basis $\{e_i^\xi\}_{i=1}^{d_\xi}$ of V_ξ . Then the family $\{\sqrt{d_\xi} \pi_{e_i^\xi, e_j^\xi}\}$ for $\xi \in \widehat{G}$, $1 \leq i, j \leq d_\xi$, is an orthonormal Hilbert basis of $L^2(G)$. In Lean: a `HilbertBasis` $(\Sigma \xi \in \widehat{G}, \text{Fin } d_\xi \times \text{Fin } d_\xi) \ (L^2(G))$.*

Proof. Combine Theorem 39 (the matrix coefficients span a dense subspace) with Theorem 42 (giving the explicit $1/d_\xi$ normalisation factor) and Theorem 41 (orthogonality across ξ). $\square$

7.2 Fourier transform and Plancherel

Definition 44 (Operator-valued Fourier transform). For $f \in L^1(G)$ and an irreducible class $[\xi]$ with representative (ρ_ξ, V_ξ) , define $\hat{f}(\xi) \in \text{End}(V_\xi)$ by $\hat{f}(\xi) := \int_G f(g) \rho_\xi(g) d\mu(g)$ (Bochner integral in the finite-dimensional space $\text{End}(V_\xi)$).

Theorem 45 (Plancherel for compact groups). For $f \in L^2(G)$: $\|f\|_{L^2(G)}^2 = \sum_{[\xi]} d_\xi \|\hat{f}(\xi)\|_{HS}^2$, where $\|\cdot\|_{HS}$ is the Hilbert–Schmidt norm on $\text{End}(V_\xi)$.

Proof. Expand f in the orthonormal basis from Corollary 43 and identify the resulting coefficients with the entries of $\hat{f}(\xi)$. $\square$

Theorem 46 (Fourier inversion). For $f \in L^2(G)$ (with L^2 convergence), $f(g) = \sum_{[\xi]} d_\xi \text{tr}(\rho_\xi(g)^{-1} \hat{f}(\xi))$.

Proof. Same expansion as in Theorem 45, summed back. $\square$

Chapter 8

Full Statement of the Peter–Weyl Theorem

The blueprint culminates in three equivalent forms of Peter–Weyl:

Theorem 47 (Peter–Weyl, Form I: representations). *Every continuous unitary representation of a compact group G on a Hilbert space decomposes (uniquely) as a Hilbert sum of finite-dimensional irreducibles. In particular, all irreducible continuous unitary representations of G are finite-dimensional.*

Proof. Apply the isotypic decomposition (Theorem 39) componentwise. Finite-dimensionality of the irreducibles follows from Corollary 28 since every irreducible can be embedded in $L^2(G)$ by the matrix-coefficient construction (or directly by the regular representation). $\square$

Theorem 48 (Peter–Weyl, Form II: matrix coefficients). *The set of matrix coefficients of finite-dimensional continuous unitary irreducible representations is uniformly dense in $C(G, \mathbb{C})$, and the suitably normalised matrix coefficients form an orthonormal Hilbert basis of $L^2(G)$.*

Proof. Combination of Theorem 35 (uniform density) and Corollary 43 (Hilbert basis). $\square$

Theorem 49 (Peter–Weyl, Form III: L^2 decomposition). *As a representation of $G \times G$ under (left, right)-translation,*

$$L^2(G) \cong \widehat{\bigoplus_{[\xi] \in \widehat{G}} V_\xi \boxtimes V_\xi^*},$$

where each summand $V_\xi \boxtimes V_\xi^$ is the finite-dimensional representation of $G \times G$ on $V_\xi \otimes V_\xi^*$ via $\rho_\xi \boxtimes \rho_\xi^*$.*

Proof. Combine Theorem 39 and Theorem 40, the latter giving the identification of each isotypic component with $V_\xi \otimes V_\xi^*$ as a $G \times G$ -representation. $\square$

Chapter 9

Mathlib-Status Summary and Roadmap

What is already in Mathlib ([ML], with \leanok). The analytic foundations are mostly there: `MeasureTheory.Measure.haarMeasure` and the `IsHaarMeasure` typeclass with all invariance, regularity, and uniqueness results (`Mathlib.MeasureTheory.Measure.Haar.{Basic, Unique}`); `MeasureTheory.Lp` E^2 μ with its inner-product structure (`Mathlib.MeasureTheory.Function.L2Space`); density of $C(G)$ in L^p on a finite regular measure space (`MeasureTheory.Lp.continuousMap_dense`); the inclusions $L^p \subseteq L^q$ on a finite measure space; the bundled `StarSubalgebra  $\mathbb{C}$   $C(G, \mathbb{C})$` type and the complex Stone–Weierstrass theorem (`ContinuousMap.starSubalgebra_topologicalClosure_eq_top_of_se`); `IsHilbertSum` and `HilbertBasis` (`Mathlib.Analysis.InnerProductSpace.l2Space`); `IsCompactOperator` as a predicate (`Mathlib.Analysis.Normed.Operator.Compact`); the multiplicative convolution `MeasureTheory.mconvolution` (`Mathlib.Analysis.Convolution`).

Trivial wrappers ([~ML]). `UnitaryRep.matrixCoeffSubalgebra_topologicalClosure_eq_top_L2` (uniform density implies L^2 density on a finite measure space).

Genuinely new contributions ([NEW]). There are roughly four blocks:

1. **Continuous unitary representation theory.** `UnitaryRep  $\mathbb{C}$   $G$   $H$` (the bundled structure) and the basic API: `UnitaryRep.Subrep`, `UnitaryRep.IsIrreducible`, `UnitaryRep.Intertwiner`, `UnitaryRep.Equiv`, the orthogonal-complement closure of invariant subspaces (Lemma 15), and the Schur lemmas (Theorem 19, Corollary 20). This is roughly the analog of what Mathlib’s `Representation` infrastructure provides in the algebraic setting, and probably belongs in a new `Mathlib.RepresentationTheory.Unitary` file (or several).
2. **Convolution as a compact self-adjoint operator on $L^2(G)$.** The chain Theorem 23, Theorem 24, Theorem 25, Lemma 29. These are essentially folklore but are not currently in Mathlib for compact non-abelian groups.
3. **The compact-operator spectral theorem (Theorem 26).** This is the most substantial gap. Mathlib has the finite-dimensional version (`LinearMap.IsSymmetric.diagonalization`) but not the infinite-dimensional compact-self-adjoint case. Formalising it is a project in its own right (perhaps even larger than the rest of this blueprint combined). A workaround might be to prove just the weaker “every compact self-adjoint operator has a non-zero

eigenvalue with finite-dimensional eigenspace”, which is enough for Corollary 28 and Theorem 33.

4. **Peter–Weyl proper.** The decomposition theorems themselves: Theorem 33 (existence of separating finite-dim irreducibles), Theorem 35 (dense matrix coefficients via Stone–Weierstrass), Theorem 39 (L^2 Hilbert sum), Theorem 40 (per-isotypic structure), Schur orthogonality, Plancherel, Fourier inversion.

Suggested formalisation order.

1. Define `UnitaryRep` and basic API (Chapter 2).
2. Convolution facts on a compact group (Chapter 4, sections 1).
3. Either (a) formalise the compact-self-adjoint spectral theorem in full generality, or (b) extract the weaker “every non-zero compact self-adjoint operator has a finite-dimensional eigenspace at a non-zero eigenvalue” — which is what we actually use — as a focused mini-project.
4. Schur’s lemma (Chapter 3).
5. The matrix-coefficient subalgebra and Stone–Weierstrass (Chapter 6, sections 1–3).
6. The Peter–Weyl decomposition and Plancherel (Chapter 6, sections 3–4 and Chapter 7).

Comparison to the finite-group blueprint. Recall that for finite G over an algebraically closed field k , the proof reduces to “Maschke + Wedderburn–Artin + algebraic closedness”, with about *four*–five genuinely new lemmas needed beyond Mathlib. For a compact (possibly infinite) topological group, the analytic infrastructure dominates and the new contributions are an order of magnitude larger; the spectral theorem alone is a sub-project of substantial size. This is in line with the historical observation that Peter–Weyl for compact groups is much harder than the finite-group case and required serious harmonic analysis to develop.